

Lecture 1B – Options and Stochastics

Dr. Fang Fang

TU Delft

- Course nr: WI4151
- Dr. Fang Fang
- Assis. Prof. of Applied Mathematics of TU Delft
- F.Fang@tudelft.nl;

Lecture 1B – Options and stochastics

- Value of call/put option and hedging
- Modeling of asset prices and randomness
- Stochastic processes for stock prices
- Markov property and efficient market hypothesis
- Ito's Lemma - definition

Options (1/2)

An option is a contract that gives the holder the right, but not the obligation, to buy or sell an underlying asset at a specified price on or before a specified date.

John C. Hull, *Options, Futures, and Other Derivatives*

- A contract between two parties
- A right, not an obligation, to the option buyer
- With fixed conditions/terms:
 - Underlying (stock price, some index, a swap contract, commodity price, FX rate, etc.)
 - Strike price
 - Exercise date(s) (sometimes can be multiple)
 - Direction (buy or sell, or “long or short”)
 - Knock-in or/and -out conditions
 - Etc.

Options (2/2)

- 26th April 1973, options were first officially traded on an exchange, The Chicago Board Options Exchange (CBOE) created standardized, listed options. Initially there were just calls on 16 stocks. Puts weren't introduced until 1977.
- Options are traded on CBOE, the American Stock Exchange, the Pacific Stock Exchange and the Philadelphia Stock Exchange.
- Currently, worldwide there are over 50 exchanges on which options are traded.

Example

- American call option¹;
 - The underlying is Apple's stock price; Maturity date is 2021-7-9; The strike price is 100\$; "Today's" price of the underlying is 143.61\$ @ 2021-7-6 9:34AM EDT; Price of the option as of "today" is 44.25\$ for selling, and 44.40\$ for buying – bid-ask spread;

Apple Inc. (AAPL)

NasdaqGS - NasdaqGS Real Time Price. Currency in USD

143.61 +1.59 (+1.12%)

As of 10:34AM EDT. Market open.

Calls for July 9, 2021

Contract Name	Last Trade Date	Strike ^	Last Price	Bid	Ask	Change	% Change	Volume	Open Interest	Implied Volatility
AAPL210709C00075000	2021-07-06 11:49AM EDT	75.00	65.80	69.35	69.45	0.00	-	77	102	429.88%
AAPL210709C00080000	2021-06-30 2:08PM EDT	80.00	56.85	64.40	64.55	0.00	-	-	6	400.00%
AAPL210709C00085000	2021-06-16 2:40PM EDT	85.00	43.90	59.30	59.45	0.00	-	40	40	355.08%
AAPL210709C00090000	2021-07-06 3:19PM EDT	90.00	52.10	54.30	54.45	0.00	-	225	986	322.07%
AAPL210709C00095000	2021-06-24 11:17AM EDT	95.00	39.15	49.25	49.40	0.00	-	3	10	286.43%
AAPL210709C00100000	2021-07-06 9:34AM EDT	100.00	42.86	44.25	44.40	0.00	-	6	36	256.64%

¹Note: this is a deep "in the money" option, which is relatively expensive. So the volume is quite low.

Option types (1/2)

- “Call” or “Put”
 - A call option gives the option holder the right to buy the underlying
 - A put option gives the option holder the right to sell the underlying
- “European”, “American” or “Bermudan”
 - An European option can be exercised only at the maturity
 - An American option can be exercised any time before the maturity
 - A Bermudan option can be exercised at a few pre-specified dates before the maturity

Option types (2/2)

- “Vanilla”, “Barrier”, “Asian”, or others
 - A vanilla option is a simple call or put option with no special features or observation dates.
 - A barrier option adds knock-in and/or knock-out features
 - An Asian option has the underlying being the average price of a basket of single-name underlyings
 - Other options also exist such as
 - Swaption, of which the underlying is a swap contract
 - Options on Futures, of which the underlying is a Futures contract
 - Etc.

Option payoffs

- The value of a European call option at the expiry T is given by:

$$V_C(T, S(T)) = \max(S(T) - K, 0).$$

- The value of a European put option at the expiry T is given by:

$$V_P(T, S(T)) = \max(K - S(T), 0).$$

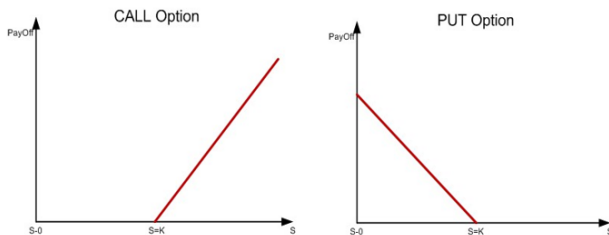


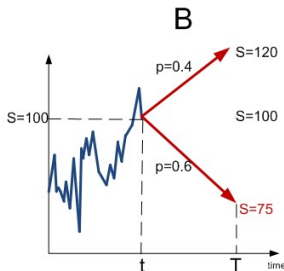
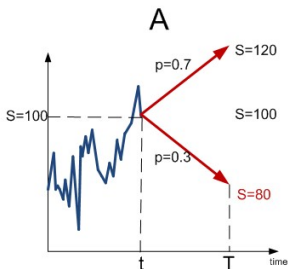
Figure: Payoff diagram for European Call (left), and European Put (right).

The value of an option

- What determines the value of an option?
 - what is the asset price today $S(t)$?
 - how long there is until expiry $T - t$?
 - how volatile is the asset $S(t)$?
- General principles:
 - The longer the time to expiry, the more time there is for the asset to rise or fall;
 - The more volatile the asset, the higher the chance that it will rise or fall;

The value of an option

- The value of an option



- For $K = 100$ which call option is more expensive

Example

- Writers, in order to hedge their short call option position, buy some (Δ) stocks, so their position equals:

$$\Pi(t) = V_{c,0} - \Delta \cdot S_0$$

- Now, we consider two scenarios:
 - Stock goes **UP**:

$$\Pi_{\text{up}} = V_{c,0} - \Delta \cdot S_0 + \Delta \cdot S_{\text{up}} - S_{\text{up}} + K \quad (1)$$

$$= V_{c,0} - \Delta \cdot S_0 + \Delta \cdot S_{\text{up}} - 20 \quad (2)$$

- Stock goes **DOWN**:

$$\Pi_{\text{dn}} = V_{c,0} - \Delta \cdot S_0 + \Delta \cdot S_{\text{dn}} \quad (3)$$

- Since writers don't want to take any risk:

$$\begin{cases} V_{c,0} - \Delta \cdot S_0 + \Delta \cdot S_{\text{up}} - 20 = 0, \\ V_{c,0} - \Delta \cdot S_0 + \Delta \cdot S_{\text{dn}} = 0. \end{cases} \quad (4)$$

Example cont.

- Simple algebra gives:

$$\Delta = \frac{20}{S_{\text{up}} - S_{\text{dn}}}.$$

- So now we have:
 - Case A:
 - $\Delta = \frac{1}{2}$, which gives $V_{c,0} = 10$.
 - Case B:
 - $\Delta = \frac{4}{9}$, which gives $V_{c,0} = 11.1$.
- How does a writer make money???

How does a writer make money? (1/2)

- From the hedging equations we imposed

$$\Pi_{\text{up}} = \Pi_{\text{dn}} = 0,$$

i.e. the terminal value of the *hedged* portfolio is zero in both states.

- This means that if the call option is sold at the price

$$V_{c,0}$$

implied by replication, the writer:

- takes *no risk*,
- but also makes *no profit*.
- In particular:
 - Case A: $\Delta = \frac{1}{2} \Rightarrow V_{c,0} = 10$,
 - Case B: $\Delta = \frac{4}{9} \Rightarrow V_{c,0} = 11.1$.
- These prices are the **no-arbitrage (fair) prices**.

How does a writer make money? (2/2)

- A writer makes money only if the option is sold for

$$V_{c,0}^{\text{sell}} > V_{c,0}^{\text{replication}}.$$

- The difference

$$V_{c,0}^{\text{sell}} - V_{c,0}^{\text{replication}}$$

is invested at the risk-free rate and becomes a *guaranteed profit* under perfect hedging.

- In practice, writers earn money due to:
 - bid–ask spreads and fees,
 - imperfect hedging (discrete rebalancing),
 - transaction costs and liquidity effects,
 - volatility and jump risk not captured by the simple model.
- **Key message:** At the fair price, a perfectly hedged writer does not make money.

Case A vs Case B

- In Case A and Case B the *possible future stock prices* (S_{up} , S_{dn}) are different, leading to different hedging ratios:

$$\Delta = \frac{20}{S_{\text{up}} - S_{\text{dn}}}.$$

- This gives:
 - Case A: $\Delta = \frac{1}{2}$, hence $V_{c,0} = 10$,
 - Case B: $\Delta = \frac{4}{9}$, hence $V_{c,0} = 11.1$.
- Although the *real-world probabilities* of the up/down moves differ between Case A and Case B, the option value is determined by **replication and no-arbitrage**, not by those probabilities.
- **Key point:** Different price dynamics $\Rightarrow$ different hedging costs $\Rightarrow$ different fair option prices.

Intrinsic value and time value of an option

- The value of an option is determined by the spot price S_t of the underlying asset, the strike K , the time to maturity $T - t$, and the uncertainty of the maturity price S_T .
- Option price = intrinsic value + time value.
- For call options:
 - Intrinsic value:

$$\max(S_t - K, 0).$$

- For put options:
 - Intrinsic value:

$$\max(K - S_t, 0).$$

- Time value is determined by the time to maturity and the uncertainty of the price of the underlying at maturity.
- Out-of-the-money options have only time value and no intrinsic value.

Option pricing theory

Two equivalent approaches:

- Hedging portfolio $\rightarrow$ Partial Differential Equation (PDE)
- Martingale approach
- Bridge between the two: Feynman-Kac theorem

Put-call parity (1/2)

- At maturity T :

$$\max(S(T) - K, 0) + K = \max(K - S(T), 0) + S(T).$$

- Equivalently,

$$C(T) + K = P(T) + S(T).$$

Interpretation

Holding a European call option and K units of cash at maturity gives the same payoff as holding a European put option and the underlying asset. By the no-arbitrage principle, the two portfolios must have the same value.

- At time t :

$$C(t) + Ke^{-r(T-t)} = P(t) + S(t).$$

Put–Call parity (2/2)

- A relation between value $V(t, S(t))$ of European call and put options with the same strike price K and expiry T . Suppose we have given two portfolios at time t :
 - Π_A : one call option plus $Ke^{-r(T-t)}$ cash (invested in the bank)
 - Π_B : one put option plus one unit of the asset
- At expiry portfolio Π_A is worth:

$$\max(S(T) - K, 0) + K = \max(S(T), K).$$

- The portfolio Π_B is worth on expiry:

$$\max(K - S(T), 0) + S(T) = \max(S(T), K).$$

- So, we conclude:

$$C(t, S(t)) + Ke^{-r(T-t)} = P(t, S(t)) + S(t).$$

- This relationship is the so-called **Put–Call Parity**.
- Note that we did not make any assumptions about stock $S(t)$!

Basic inequalities for option values (1/2)

- The no-arbitrage assumption also implies that if one portfolio always pays more than another portfolio, it must be more expensive.
- The portfolio of holding a European call option with strike K and $Ke^{-r(T-t)}$ in the deposit account from time t is worth on expiry $\max(S(T), K)$. This time- T value is greater than the value of one unit of the underlying asset $S(T)$.
- Therefore, the total price of the call and the deposit has to be greater than the price of the underlying:

$$C(t) \geq S(t) - Ke^{-r(T-t)}, \quad t \leq T.$$

- Because a call cannot take negative value, we can sharpen the lower bound as

$$C(t) \geq \max(S(t) - Ke^{-r(T-t)}, 0).$$

- A call, on the other hand, is always cheaper than the underlying asset, i.e. $C(t) \leq S(t)$.

Arbitrage

- There is never an opportunity to make a risk-free profit that gives a greater return than that provided by a bank.
- **Example:** Suppose we have given a portfolio:

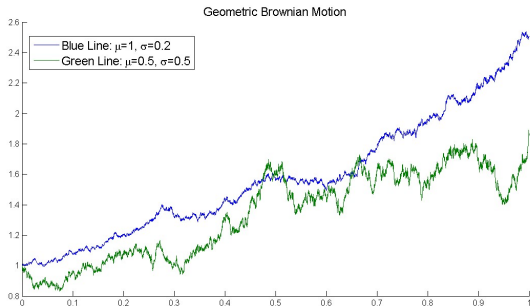
$$\Pi = V_P(t, S(t)) - V_C(t, S(t)) + S(t),$$

with $V_P(t, S(t))$ and $V_C(t, S(t))$ having expiry date T , and strike K .

- **Question:** What is the “fair” price of this portfolio?

Randomness

- Asset prices thus have a large element of randomness. This does not mean that we cannot model these prices, but it does mean that the modelling must be done in a probabilistic sense.
- A first model for asset prices, a stochastic differential equation (SDE), is geometric Brownian motion.
- We will also look in detail into so-called stochastic volatility SDE models, like the Heston dynamics.



Asset price data (1/3)

- When investing, the main concern is that the **return** on the investment is satisfactory. Suppose we have given asset S_t , then

$$\text{Return} = \frac{\text{Stock tomorrow} - \text{Stock today}}{\text{Stock today}} = \frac{S(t + \Delta) - S(t)}{S(t)}.$$

- Let's see it in practice! We take the S&P index from **30-04-2002** to **09-11-2007**, daily monitored.

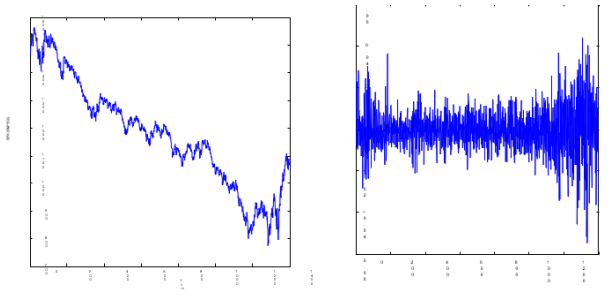
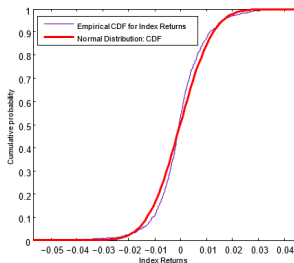
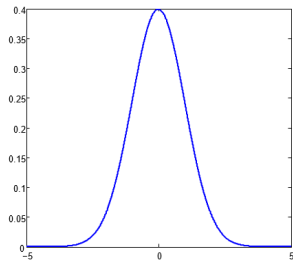


Figure: Left: S&P Index, Right: Index Return

Asset price data (2/3)

- From the data in this example we find that the mean is -0.000172 and the standard deviation is 0.0121 ;
- The distribution of the returns has been scaled and translated to give it a mean of zero, a standard deviation of one and an area under the curve of one.



Asset price data (3/3)

- Below is the plot from Jan-13-2020 till Aug-13-2021, covering the Covid pandemic from its beginning till the time of writing.



Randomness of the stock prices

- Daily returns for a certain asset look like noise!
- We can model the noise!

Definition (Stochastic Process)

- A *stochastic process* $X(t)$ is a collection of random variables

$$(X(t), t \in \mathcal{T}) = (X(t)(\omega), t \in \mathcal{T}, \omega \in \Omega).$$

- A stochastic process $X(t)$ is a function of two variables:
 - For a fixed instant t , it is a random variable:

$$X(t) = X(t)(t \in \mathcal{T}), \quad \omega \in \Omega.$$

- For a fixed random outcome $\omega \in \Omega$, it is a function of time:

$$X(t) = X(t)(\omega), \quad t \in \mathcal{T}.$$

Definition (Wiener process)

- A stochastic process $W(t)$ for $t \in [0, \infty)$ is called a *Wiener process* (or *Brownian motion*) if the following conditions are satisfied:

- It starts at zero:

$$W(0) = 0.$$

- It has stationary, independent increments.
- For every $t > 0$, $W(t)$ has a normal distribution with mean 0 and variance t :

$$W(t) \sim \mathcal{N}(0, t).$$

- It has almost surely continuous paths, with **no jumps**.

Process trajectories of Wiener process

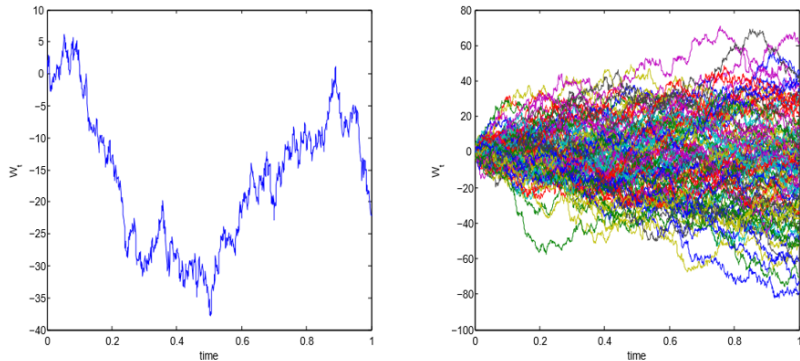


Figure: Sample paths of Brownian motion on $[0,1]$. Left: 1 path, Right: 100 paths.

Modeling the stock prices/returns: Samuelson's model

- The most popular stochastic process for generating stock prices is the *Geometric Brownian Motion* (GBM):

$$dS(t) = \mu S(t) dt + \sigma S(t) dW(t).$$

where:

- μdt is the deterministic return (drift),
- $\sigma dW(t)$ is the random change, with $dW(t)$ normally distributed with mean 0 and variance dt ,
- the SDE is the short form for

$$S(t) = S_0 + \int_{t_0}^t \bar{\mu}(s, S(s)) ds + \int_{t_0}^t \bar{\sigma}(s, S(s)) dW(s),$$

with $\bar{\mu}(t, S(t)) = \mu S(t)$ the drift and $\bar{\sigma}(t, S(t)) = \sigma S(t)$ the diffusion term.

- This can be translated (in discrete time) to:

$$S(t + \Delta t) = S(t) + \mu S(t) \Delta t + \sigma S(t) (W(t + \Delta t) - W(t)).$$

Samuelson's model

- Let us check the moments of $dS(t)$:
 - For the expectation we find:

$$\begin{aligned}\mathbb{E}_t[dS(t)] &= \mu \mathbb{E}_t[S(t) dt] + \sigma \mathbb{E}_t[S(t) dW(t)] \\ &= \mu S(t) dt.\end{aligned}$$

- The variance is equal to

$$\begin{aligned}\text{Var}_t(dS(t)) &= \mathbb{E}_t[dS^2(t)] - (\mathbb{E}_t[dS(t)])^2 \\ &= \sigma^2 S^2(t) dt.\end{aligned}$$

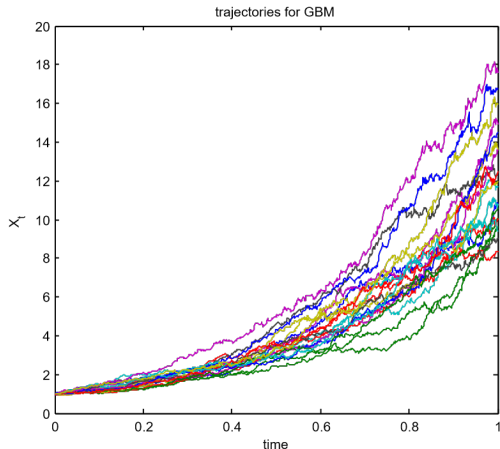
- The equation does not refer to past history of the asset price.
- Independence from the past is also known as the **Markov property**.

Properties

- This model fits real time data very well for equities and indices. (The agreement with currencies is less good, especially in the long term.)
- Discrepancies: real data appears to have a greater probability of large rises and falls than the model predicts.
- The drift is not apparent over short time scales for which the volatility dominates.
- Over long time scales, for instance decades, the drift becomes important.
- Alternatives: asset price model includes jumps or adopt stochastic volatility.

Modeling the returns: continuation

$$S(t + \Delta t) = S(t) + \mu S(t) \Delta t + \sigma S(t) (W(t + \Delta t) - W(t)).$$



But one second:

- What is the economical meaning of Markov property?
- Do we observe it in the data?

Efficient market hypotheses (1/3)

- In Fama (1970)'s paper, he forms three efficient market hypotheses: "Efficient Capital Markets: A Review of Theory and Empirical Work". Journal of Finance. 25 (2): 383–417
- <http://efinance.org.cn/cn/fm/Efficient>

THE PRIMARY ROLE of the capital market is allocation of ownership of the economy's capital stock. In general terms, the ideal is a market in which prices provide accurate signals for resource allocation: that is, a market in which firms can make production-investment decisions, and investors can choose among the securities that represent ownership of firms' activities under the assumption that security prices at any time "fully reflect" all available information. A market in which prices always "fully reflect" available information is called "efficient."

Efficient market hypotheses (2/3)

This paper reviews the theoretical and empirical literature on the efficient markets model. After a discussion of the theory, empirical work concerned with the adjustment of security prices to three relevant information subsets is considered. First, *weak form* tests, in which the information set is just historical prices, are discussed. Then *semi-strong form* tests, in which the con-

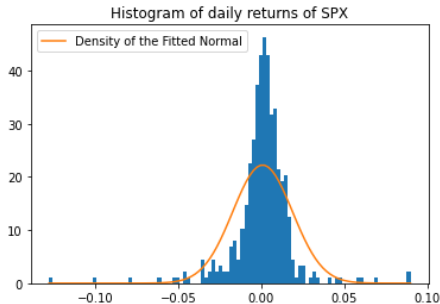
- The weak form of the efficient market hypothesis:
- the current asset price reflects all past price information.
- The weak efficient market hypothesis implies that, if we want to predict the asset price at some future time, knowing the complete history of the asset price gives no advantage over just knowing its current price.

Efficient market hypotheses (3/3)

- The semi-strong form states that because all the public information (other than the price history) is already contained in the current price, investors cannot utilize either technical or fundamental analysis; the private information is not priced and therefore knowing private information is still an advantage.
- The strong form states all the price history, other public and private information, are already reflected in the current price of the asset. Knowing private information is no more useful.

is considered. First, *weak form* tests, in which the information set is just historical prices, are discussed. Then *semi-strong form* tests, in which the concern is whether prices efficiently adjust to other information that is obviously publicly available (e.g., announcements of annual earnings, stock splits, etc.) are considered. Finally, *strong form* tests concerned with whether given investors or groups have monopolistic access to any information relevant for price formation are reviewed.¹ We shall conclude that, with but a few exceptions, the efficient markets model stands up well.

Test for i.i.d. and normality (1/3)



- If the returns are an i.i.d. sequence, the incremental changes of asset prices are independent. It further implies the weak form of efficient market hypothesis.
- We test whether the daily log-returns of the SPX index in the sample period are distributed as i.i.d. normal random variables.
- First we visualize the distribution of log-returns. The plot shows the empirical distribution is approximately symmetric around 0.

Test for i.i.d. and normality (2/3)

- Given the daily log-returns $\{r_t\}$, the sample mean and sample variance are:

$$\mu_r = \frac{1}{T} \sum r_t \approx 0.000776,$$

$$\sigma_r^2 = \frac{1}{T-1} \sum (r_t - \mu_r)^2 \approx 0.000322.$$

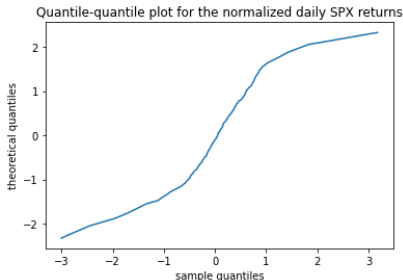
- The assumptions of i.i.d. and normality imply

$$r_t \sim \mathcal{N}(\mu_r, \sigma_r^2).$$

- We can calculate the normalized returns. The normalized returns are supposed to follow the standard normal distribution $\mathcal{N}(0, 1)$ by the properties of the normal distribution, i.e.

$$\hat{r}_t = \frac{r_t - \mu_r}{\sqrt{\sigma_r^2}} \sim \mathcal{N}(0, 1).$$

Test of i.i.d. and normality (3/3)



- The QQ plot suggests that the assumption of i.i.d. normality are not accurate at large quantiles.
- The range of theoretical quantiles (about $[-2.3, 2.3]$) is narrower than the sample quantiles (about $[-3, 3]$). It suggests that the empirical distribution is wider than the normal distribution.
- The reason is that in the data there are more large positive and negative returns, relative to a normal distribution. It is called the fat tail behavior.

Remarks

- The weak form of the efficient market hypothesis and the assumption of i.i.d. for asset returns are *not* equivalent.
- The weak form of the efficient market hypothesis can be formulated mathematically via the **Markov property**:

$$\mathbb{P}(S_{t+1} \leq \bar{S} \mid S_t) = \mathbb{P}(S_{t+1} \leq \bar{S} \mid S_t, S_{t-1}, S_{t-2}, \dots).$$

- $\mathbb{P}(S_{t+1} \leq \bar{S} \mid S_t)$ is the conditional probability distribution of the asset price at time $t + 1$, given the price at time t .
 - $\mathbb{P}(S_{t+1} \leq \bar{S} \mid S_t, S_{t-1}, S_{t-2}, \dots)$ is the conditional probability distribution of the asset price at time $t + 1$, given all historical prices.
- If the assumption of i.i.d. returns is valid, it implies the weak form of the efficient market hypothesis because of the independence of returns, but not vice versa.

Types of processes

- We compare the behaviour of the following SDEs:
 - **Arithmetic Brownian motion (ABM):**

$$dX(t) = \mu dt + \sigma dW(t).$$

- **Geometric Brownian Motion (GBM):**

$$dX(t) = \mu X(t) dt + \sigma X(t) dW(t).$$

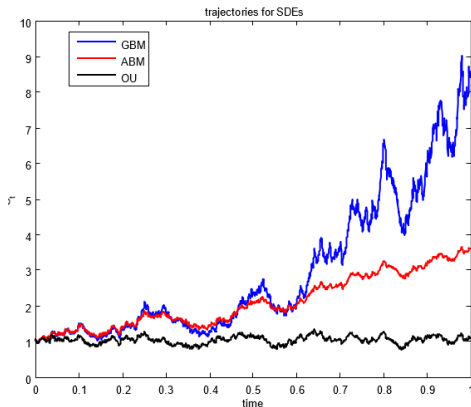
- **Ornstein–Uhlenbeck mean-reverting process (OU):**

$$dX(t) = \kappa(\theta - X(t)) dt + \sigma dW(t).$$

where μ , σ , κ and θ are known constants, and $W(t)$ is a Wiener process.

- We will also cover multidimensional extensions.
- What is (are) the difference(s) between these processes?
- What property do they share?

Process trajectories



- Figure: Process trajectories generated from the same random path. The parameters are: $\mu = 0.05$, $\sigma = 0.7$, $\kappa = 1.5$ and $\theta = 1$.

Ito's Lemma

- Ito's Lemma is fundamental for stochastic processes. It helps in deriving solutions to stochastic differential equations (SDEs).
- Lemma (Ito's Lemma).

Suppose $X(t)$ follows an Ito process:

$$dX(t) = \bar{\mu}(t, X(t)) dt + \bar{\sigma}(t, X(t)) dW(t), \quad X_0 = x_0,$$

where $\bar{\mu}(t, X(t))$ is the drift term and $\bar{\sigma}(t, X(t))$ is the diffusion term.

Let $g(t, X(t))$ be a function with continuous derivatives up to second order. Then

$$Y(t) := g(t, X(t))$$

follows an Ito process with the same Wiener process $W(t)$:

$$dY(t) = \left(\frac{\partial g}{\partial t} + \bar{\mu} \frac{\partial g}{\partial x} + \frac{1}{2} \bar{\sigma}^2 \frac{\partial^2 g}{\partial x^2} \right) dt + \bar{\sigma} \frac{\partial g}{\partial x} dW(t).$$